Blida Site Health Scoring Methodology

This document outlines the mathematical models and thresholds required to calculate real-time Health Scores (0-100) for your rooms, devices, and the overall site. This methodology is designed to be directly implemented into a dashboard backend (e.g., Python/Pandas, SQL, or Grafana).

1. The Scoring Scale

All components use a standard 0-100 scale:

- **90 - 100 (Optimal):** Operating perfectly within expected parameters.
- **70 - 89 (Warning):** Sub-optimal conditions. Requires monitoring or scheduled maintenance.
- **0 - 69 (Critical):** Immediate action required. High risk of hardware degradation or failure.

2. Room Health Score (Environmental)

Applies to: Switch Room, Battery Room, ENR Room, V-SAT Room.
Inputs Required: `Temperature (°C)` (`DS2_SWITCH MSC10`, `DS2_BAT MSC10`, `DS2_ENR MSC 10`, `DS2_V-SAT MSC10`). *Note: Humidity is excluded as it is not present in the simulated dataset.*

A. Temperature Component (100% Weight)

The score scales dynamically based on a customizable "High Threshold" (T_{max}) defined per room (e.g., $T_{max} = 25°C$ for the Battery Room, $T_{max} = 30°C$ for the Switch Room).

- **Ideal Range ($< 80\%$ of T_{max}): $\rightarrow$ 100 Points**
- **Warning Range (80% to 100% of T_{max}):**
  - *Formula:* $100 - \left(\frac{T_{current} - (0.8 \times T_{max})}{0.2 \times T_{max}} \times 30\right)$
  - *(Score drops linearly from 99 to 70)*
- **Critical Range ($> 100\%$ of T_{max}):**
  - *Formula:* $70 - \left(\frac{T_{current} - T_{max}}{0.1 \times T_{max}} \times 70\right)$
  - *(Score drops from 69 to 0, floored at 0. Zero is hit if temp exceeds the threshold by 10%)*

Final Room Score: `Temp Score`

3. UPS Device Health Score (Power)

Inputs Required: `Output Load (%) L1/L2/L3` (`DS3_Output_Load_L1`, etc.), `Battery Capacity (%)` (`DS3_Battery_Capacity`), `UPS Temperature (°C)` (`DS3_UPS_Temp`), `Alarms` (`DS3_UPS_Alarm`, `DS3_Clim_Alarm`, `DS2_SCADA_Alarm`).

The UPS score is calculated from a base score, minus active penalties.

A. Base Metrics

1. **Capacity Load Score (35% Weight)**
   - *Logic:* A UPS running at 95% capacity is highly stressed and leaves no room for failover.
   - *Calculation:* Use the Maximum Load across L1, L2, L3.
   - $<$ 60% Load = **100 Points**
   - 60% - 85% Load = **Linear scale down to 70 Points**
   - $>$ 85% Load = **Linear scale down to 0 Points**
2. **Phase Balance Score (20% Weight)**
   - *Logic:* The load should be distributed evenly across L1, L2, and L3. Unbalanced phases stress the inverter.
   - *Calculation:* Unbalance % = `Max(L1, L2, L3) - Min(L1, L2, L3)`
   - $<$ 10% Unbalance = **100 Points**
   - 10% - 25% Unbalance = **70 Points**
   - $>$ 25% Unbalance = **30 Points**
3. **Battery Float Health (30% Weight)**
   - *Logic:* Assuming grid power is available, the battery should be floating at 100%. If it is dropping while the grid is connected, the battery cells are failing.
   - *Calculation (Grid ON):* Battery at 100% = **100 Points**. Below 95% = **0 Points** (Critical fault).
   - *(Note: If Grid is OFF, this metric is ignored and the weight is redistributed to the other metrics, as battery drop is expected).*
4. **Internal Temperature (15% Weight)**
   - Similar to room temperature scoring, but for the UPS's internal sensor. Ideal $<$ 30°C.

B. Alarm Deductions (The Penalty System)

Subtract fixed points from the Final Base Score based on active alarms detected in the event datasets. Here is the precise classification of alarms from the logs:

1. Information Events (-0 Points)

Logs state changes that do not indicate a failure. Just for auditing.

- **UPS Log (`PageLogEvent`):** `[Alarm Name] has been restored` (e.g., "General Alarm has been restored"), `Load protected by Inverter`
- **Clim Log (`eventlog clim`):** `System started`, `admin login`, `manual logout`, `Alarm reset`

2. Warning Alarms (-20 Points)

Sub-optimal states that require attention but haven't caused a full outage.

- **UPS Log (`PageLogEvent`):** `Operating on Battery` (Grid lost, but UPS is surviving), `Charger Preventive Alarm`
- **SCADA Log (`ALARMES SCADA`):** `ALARME CLIM LIEBERT` (AC Warning), `UPS [X] ON BYPASS` (Running on raw grid power without inverter protection)
- **Clim Log (`eventlog clim`):** `wrong password`, `Alarm set` (User triggered alarm limits)

3. Critical Alarms (-60 Points)

Major failures indicating equipment damage, imminent shutdown, or critical environment breaches.

- **UPS Log (`PageLogEvent`):** `General Alarm`, `Unit [X] General Alarm`, `Rectifier Input Supply not OK`, `Bypass Input Supply not OK`
- **SCADA Log (`ALARMES SCADA`):** `Absence de Tension Reseau` (Total grid blackout), `TEMPERATURE HAUTE` (High Temp Critical), `UPS [X] FAILURE` (Complete UPS failure)

Final UPS Score: `(Weighted Base Metrics) - Sum(Alarm Deductions)` (Floored at 0)

4. Overall Site Health Score (Blida)

The overall site score is not a simple average. It is a **Weighted Average** that reflects the criticality of the infrastructure. Power is the most critical, followed by the battery environment.

- **UPS Health Score:** 40% Weight *(If power fails, everything fails)*
- **Battery Room Health:** 25% Weight *(Thermal runaway degrades backup power)*
- **Switch Room Health:** 15% Weight *(IT equipment cooling)*
- **ENR Room Health:** 10% Weight
- **V-SAT Room Health:** 10% Weight

Critical Site Overrides (The "Kill Switch")

Regardless of the weighted average, the dashboard should force the Overall Site Score to specific values under extreme conditions:

1. **Grid Power Failure:** If `DS3_Input_Voltage_L1`, `L2`, `L3 == 0` for all 3 phases, cap the Maximum Site Score at **50**. (The site is alive, but in a highly vulnerable state).
2. **Imminent Shutdown:** If Grid Power is 0 AND `DS3_Battery_Capacity` $< 20\%$, force Site Score to **5** (Code Red).